

The rise of teamwork and career prospects in academic science



The rise in team size in academic science has generated an unintended side effect: junior scientists are less likely to secure research funding or obtain tenure and are more likely to leave academia.

Career prospects for young people pursuing a career in academic science are increasingly grim^{1–5}. Tenure-track positions have stagnated relative to non-tenure-track positions^{6,7}. The probability of obtaining a tenured position and enough funding for a laboratory of one's own has declined sharply⁸. In 1998, a commission of the National Research Council described a “growing crisis in expectation that grips young life scientists who face difficulty achieving their career objectives”⁹, and the crisis has worsened and spread to other fields since then^{5,6}.

It has proved difficult to identify the causes of this broad-based and long-term trend. Some point to policy changes such as the elimination of mandatory retirement, which cause senior academics to remain in their roles for longer, leaving fewer openings for young academics to be promoted^{10–12}.

At the same time that career prospects have worsened, science has become more specialized and also more collaborative¹³. Research teams have grown larger^{13,14}; as an indicator, the average number of authors per paper has increased from 2.12 in 1970 to 4.06 in 2004 (ref. 15).

Although the prevalence and impact of teams on science contributions and productivity have been investigated in the literature, no studies have examined the impact of increasing team sizes on careers. Interestingly, theoretical work by de Fontenay et al.¹⁶ suggests that if team sizes increase in a field, young scientists' careers may suffer. When scientists work in larger teams, the contribution of each team member becomes more difficult to assess, and it becomes more difficult for universities and funding agencies to identify promising junior scientists. The normal information mechanisms of personal recommendation from a laboratory head, and the quality of one's resume, still carry considerable information; but they probably carry

less information when teams are large. For example, whereas in some fields the order of authorship indicates the time contribution of different authors, it may not indicate which author had the key insights that determined the paper's quality. And a scientist heading a very large laboratory will have less information on the quality of each scientist working in the laboratory. Consequently, more of the available funding may shift toward senior scientists. Furthermore, the ‘Matthew effect’¹⁷ could mean that young scientists' achievements are attributed to senior members of their team, causing more rewards to flow to senior scientists.

We pose the empirical question: how much of the decline in young academics' career prospects is due to increasing team size in their field, and how much is due to other factors? We run a regression on career outcomes at the individual level to control for any changes in the characteristics of young scientists (such as whether the scientist obtained their PhD from a top-ranked school). The team size regressor is average team size in one's field at time of graduation.

It is natural to ask why we are not using individual team size (the size of the teams that the individual has been part of) in that regression. Doing so would give the wrong answer. When rewards are for relative performance rather than absolute performance, measuring individual-level variables can be misleading. For example, consider a national exam that is graded on a bell curve, or that ranks students relative to each other. Any regression of individual-level hours studied on individual performance in the exam will show a strong positive effect; but it would be obviously incorrect to infer that if all students studied more, their performance would improve relative to one another. A multi-year regression of individual performance on the average hours that students studied in their exam year would show no effect.

The analogy to bell curves is an apt one in this case. A regression of an individual's career success on the size of teams she published with would be likely to show a strong positive effect. Having more co-authors significantly raises citations and the likelihood of receiving funding¹⁸. Teamwork leads to higher academic productivity in science¹⁹, medicine²⁰ and social science^{21–23}, with obvious benefits for one's career. The literature has shown that smaller teams tend to disrupt science and larger teams are associated with the development of existing ideas²⁴. Probing this result further, the organization of teams also matters: flat teams innovate whereas hierarchical teams develop²⁵. Multi-university teams are growing in prevalence and produce higher-impact papers²⁶. But rewards in science are largely about relative—not absolute—performance within one's field. Therefore, the impact of an individual's team size on productivity will not be included in a regression of career success on average team size in the field. But the regression will still pick up the ‘noisy signals’ effect whereby young scientists' abilities are harder to determine relative to those of established senior scientists in their field. Thus, we test the theoretical prediction that an increase in average team sizes in a field has a negative impact on young scientists in that field¹⁶.

Results

Our analysis examines academic career outcomes of US-trained PhD graduates in science, engineering and health (SEH) fields working in the United States. We use nationally representative, longitudinal, biennial data from the Survey of Doctorate Recipients (SDR) from the National Science Foundation (NSF), which tracks PhD graduates from PhD completion until age 76. Our focus will be on understanding the association between the average team size at the time of graduation and subsequent career outcomes. Using the SDR has the

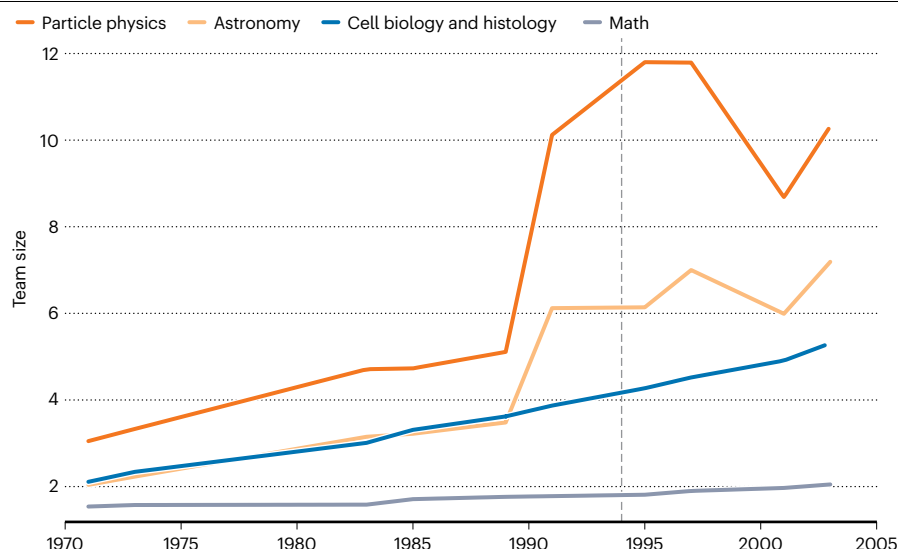


Fig. 1 | Evolution of team size in selected PhD fields, 1970–2004. Team size did not evolve smoothly across fields.

advantage of covering career outcomes for all scientists. Studies that identify individual teams based on the subset of scientists who continue to write research papers will, by definition, miss most of those who leave academic careers. Thus, using publication data to study the ‘noisy signal’ effect would produce biased results.

Sample. Our sample consists of 10 cohorts of SDR respondents who earned a PhD in 217 SEH fields that are part of the seven broad fields of life science, physical science, social science, health, psychology, math and computer science, and engineering. Our survival analysis follows these individuals from their graduation (specifically, from the survey date within three years of their graduation) through to 2013, using 18 waves of SDR data collected between 1973 and 2013. Hence, the first cohort in our data, which consists of individuals who graduated in 1969, is followed for up to 44 years from 1973 to 2013, and the last graduating cohort is followed for up to 9 years from 2004 to 2013. In our main analysis, we further restrict the sample to PhD graduates who started in academia (those who responded that they were in academia at any time in the first six years after their doctorate).

Our measure of team size is based on publication data from the core collection of Web of Science (WoS). We used the data from the Science Index, covering over 8,850 journals across 179 subject categories, and the Social Sciences Index, covering over 3,200 journals across 56 categories. Since team size evolves

slowly over time, we measure team size at 11 intervals during the years from 1970 to 2004. To combine the SDR individual-level data with the WoS team size data, we created a cross-walk between PhD fields in the SDR data using the Survey of Earned Doctorates (SED) classification and subject categories in the Web of Science data (see Supplementary Note).

Following Wuchty et al.¹⁵, we measure team size as the weighted average of the number of authors in all papers published in the individual’s PhD field f circa the graduation year y (between 1969 and 2004). We use weights because some PhD fields were matched to more than one WoS subject category. The weights are the proportion of papers in each subject category that the PhD field was matched with. On average, team size was 1.8 in 1970 and increased to 3.6 in 2004. Supplementary Table 1 reports the changes in team size by PhD field, ordered by broad field of science, and Fig. 1 shows the evolution in selected broad fields. Importantly for the predictive power of our estimation, team size did not evolve smoothly across fields. Data on career outcomes from the SDR confirm that outcomes deteriorated over the same period that team size increased.

Tenure-track positions. Junior scientists hope to secure a tenure-track position to run their own lab^{1–5}. Figure 2, which reports on outcomes 10–12 years after the doctorate, compares those who graduated at the beginning of our sample, 1969–1974 (which we refer to

as the 70s), with those graduating at the end of our sample, in 1995–2004 (which we refer to as the 90s). The figure shows a decline in the share of scientists employed in academic tenure stream positions in every science field with the exception of health. The decline was most evident in the life sciences, with a 15% decrease.

Many of the tenure-track positions have been replaced by academic non-tenure-track jobs (for example, as postdocs, lecturers and research scientists). For all science fields, the share of all academics who were employed in tenure-track roles 10–12 years after graduation fell from 39.6% to 27.1% while the share employed in academic non-tenure-track roles increased from 7.4% to 14.4% between the 70s and the 90s.

Tenure. Achieving tenure is a milestone in an academic career as it is a near-guarantee of professional stability⁴. Supplementary Table 2 shows that the share of all academics with tenure declined from 53% in 1997 to 47% in 2013, and the percentage of early-career researchers (defined as doctorate holders 7–10 years from graduation) with tenure decreased from 37% in 1997 to 27% in 2013. The only exception is in computer sciences, where the share of academics with tenure increased from 45.5% in 1997 to 57.1% in 2013.

Research funding. There is a close connection between the decline in tenure-track and tenured positions and the decline in funding success. The SDR does not include data on whether a scientist receives any grants as principal investigator (PI), but asks respondents whether any of their work during the previous year was supported by contracts or grants from the US government. We use this information to construct a proxy for receiving a grant as PI by interacting ‘receiving government support’ with having a tenure-track job.

As Supplementary Fig. 1 shows, the average age of funded scientists has increased over time, from people in their 30s in the 1970s to over 50, in most science fields. This increase of more than a decade cannot be driven by longer completion times for PhDs, which have not increased by more than 1 year in the SDR. This is similar to the trend for US National Institutes of Health (NIH)-funded scientists^{10,27} (Supplementary Fig. 2).

Exiting academia. The lack of career prospects in academia may induce young researchers to leave academia. We estimated the share and counts, by field, of those who had exited

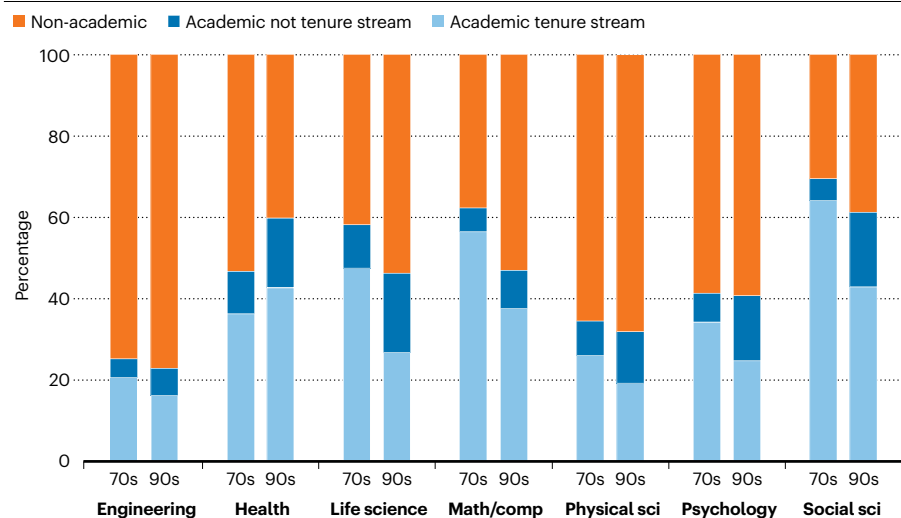


Fig. 2 | Percentage of doctorates by employment sector, 1969–1974 (70s) and 1995–2004 (90s). Academic tenure stream includes tenure track and tenured faculty. Academic non-tenure stream includes postdocs and lecturers. Non-academic includes industry, government, non-profit sector and other employment.

from academia 10–12 years after the completion of their doctorate in Supplementary Fig. 3 for doctorate holders who graduated in 1973 and 2003. Exit rates have increased strongly in life sciences but have decreased slightly in some fields.

Using SDR data at the individual level, we estimate whether the team size in a student's field at the time of graduation affects the probability that (i) the individual was employed in a tenure-track or tenured academic position; (ii) the individual was employed in a tenured academic position; (iii) the individual had exited academia; (iv) the individual worked in a field outside of their science degree (as a proxy for exiting science); and (v) the individual was a PI and supported by contracts or grants from the US government (as a proxy for whether the individual receives research funding as a PI). We estimate both probit models and Cox proportional hazard models for these outcomes.

Our models include controls for demographic characteristics such as gender, race, marital status, number of children and foreign-born status. We interact field team size with females and foreign-born, allowing for the possibility that junior scientists from those demographic groups are more likely to be affected by field team size. We control for the quality of the university from which the scientist graduated by matching the university's rank in terms of NSF funding within that PhD field and year. We also control for graduation year fixed effects to allow for the possibility that both team size and tenure rates have

increased over time for exogenous reasons. Finally, we include measures of broad field of study within each cohort, to control for potential differences in the demand for scientific fields over time. Supplementary Table 3 shows that the probability of the first job being on the tenure track decreased and the probability of first job being a postdoc increased with team size.

Impact of team size on careers

Figure 3 graphs the hazard ratio (HR) of the association between field team size and outcomes of interest, with pointwise 95% confidence intervals. Estimation results appear in Supplementary Tables 4–8. One can interpret $(1 - \text{HR}) \times 100$ as the percentage change in the outcome that is associated with a one-author increase in team size. The estimates in Fig. 3 suggest that having one more author per paper in an individual's PhD field is associated with a $(1 - 0.75) \times 100 = 25\%$ lower probability of obtaining a tenure-track appointment ($P < 0.05$), a 28% reduction in the likelihood of obtaining tenure ($P < 0.05$) and an 11% reduction in the likelihood of obtaining federal research funding ($P < 0.05$). Given the importance of tenure as a career milestone and the stiff competition among academics for funding, these are important effects. Given that field team size has increased by 1.8 authors from 1970 to 2004, the implication is that the increase in field team size more than explains the average decline in tenure prospects that graduates have experienced over this period.

There is no statistically significant effect of field team size on the probability of exiting academia, at least for men. This indicates that people are either taking longer post-doctoral appointments or ending up in non-tenure-track positions. However, team size is associated with a 6% increase in the probability of exiting science entirely ($P < 0.05$), measured by taking a job outside of one's scientific field.

We repeat this analysis using the entire sample of science and social science doctorates, meaning that we do not restrict the sample to individuals who started their careers in the academic sector. We do not estimate the probability of exiting academia because people in this sample do not necessarily start their careers in academia. As shown in Supplementary Table 9, one more author per team in a scientist's field implies that individuals are 24.3% less likely to hold a tenure-track (or tenured) job, 29.1% less likely to receive tenure, 11.4% less likely to receive federal funding and 10.7% more likely to leave science (all estimates $P < 0.01$). As with the academics-only sample, increasing average team size can more than account for the observed average decline in tenure prospects.

Impact of team size on women and the foreign-born. If output is published by larger teams, the publication record contains less information on the individual's ability, particularly for younger, unproven individuals. It is rational for funding agencies to channel more funding toward established scientists, in response¹⁶.

Poor information also raises the likelihood of bias, as the inferences drawn about an individual scientist's quality may be affected by the decision-maker's priors. There is the bias in favor of established scientists (the Matthew effect¹⁷), in which a high-quality paper tends to be attributed to the best-known scientist on the project. There are also traditional biases that can be detrimental to women and to foreign-born scientists. The evidence about how important such biases are in funding and promotion processes is mixed^{28,29}.

Our hypothesis would suggest that increasing team sizes would be particularly harmful to women scientists and to foreign-born scientists as there is more risk of bias toward those demographic groups, in the absence of reliable information. The data confirm that increasing team sizes have been more harmful to women and the foreign-born. Supplementary Tables 4–8 provide the full set of results for the Cox hazard model. They indicate that the presence of one more

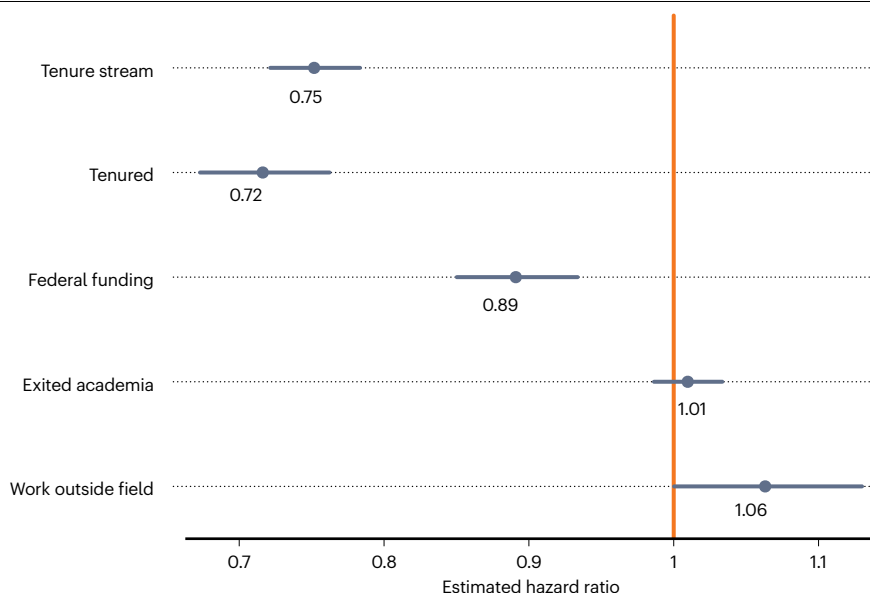


Fig. 3 | Hazard ratio estimates of the effect of team size on career outcomes. Estimates using 1973–2004 Survey of Doctorate Recipients. Each hazard ratio is estimated by a separate model for each outcome and shown with 95% confidence intervals.

author per paper lowers the likelihood that a woman holds a tenure-track or tenured job by a further 5.6% ($P < 0.05$), in addition to the 24.8% lower likelihood for all scientists. We do not find a significantly lower probability of women getting tenure, however. Women are also 5.5% ($P < 0.05$) less likely to receive federal funding and 6.4% ($P < 0.05$) more likely to leave academia. Foreign-born scientists are 5.1% less likely to hold a tenure-track or tenured appointment, 8% less likely to obtain tenure and 6.7% less likely to receive funding (all at the $P < 0.05$) if there is one more author per paper in their field; again, this is in addition to the effects for all scientists described above. We do not find a significantly larger likelihood that foreign-born will exit academia, however.

Alternative explanations for the effect of team size. It is important to account for major changes in policy to see if they are driving our results, with the most notable policy being the end of mandatory retirement for academics in 1994. We also consider the impact of the doubling of the NIH budget between 1998 and 2003³⁰.

It is possible that our team size effects are simply the manifestation of mandatory retirement; however, our evidence indicates that this is unlikely. Figure 1 shows no change in the trend in team size post 1994. We analyze this further in Supplementary Table 10, which

shows the percentage change in team size from 1970 to 1992 and from 1992 to 2004 by broad field of science. Overall, team size grew by over 46% from the 1970s to the early 1990s, whereas it grew by 24% after the end of mandatory retirement. In the fields of computer science, social science and health, team size grew more rapidly after the end of mandatory retirement. However, in all remaining fields, much of the growth in team size occurred before that policy change.

We estimated the hazard model separately for the sub-samples of individuals who graduated before and after the mandatory retirement policy change in Supplementary Table 11. For all outcomes, the magnitude of the team size estimate is larger before the mandatory retirement change than afterwards. Figure 4 visually demonstrates that the slope of the relationship between team size and exit rates was flatter after the change. These results suggest that the end of mandatory retirement might have exacerbated the career crisis (by increasing the likelihood that graduates will leave academia and reducing the likelihood that they obtain a tenure-track job), but it does not explain the team size effect that we find.

Next, we examine the impact of the NIH doubling on overall team size. Between 1998 and 2003, the NIH budget doubled in real terms³⁰. However, funding for other science fields remained constant. Much of the

NIH budget funds basic biomedical sciences. The NIH doubling could have reduced team size if more researchers had been funded during that time. We test whether team size was smaller in cohorts graduating in basic biomedical fields during the years of the NIH doubling (1998–2003) and with a three-year lag (2000–2006). We graph the team size coefficients from these difference-in-difference regressions, comparing biomedical fields to all science fields and to life science and chemistry fields in Supplementary Fig. 4. In our preferred specification, where the effect of the NIH doubling was lagged by three years, team size in basic biomedical fields increased by 0.55 ($P < 0.001$) people relative to all other science fields. When we compare to only life science fields and chemistry, team size increased in basic biomedical fields by nearly 0.42 people ($P < 0.012$). These results show that the doubling of the NIH budget was indeed associated with an increase in team size in biomedical fields.

Discussion

Our evidence suggests that increases in average team size worsen the career prospects of scientists. Following 10 cohorts of science and social science graduates over time, we found that as team size at the time of graduation increased across fields and time, individuals were less likely to obtain tenure-track jobs, receive tenure and receive federal research funding. In addition, these individuals were more likely to leave academia and their scientific field. Surprisingly, the estimated impact of team size is able to account for the entire decline in tenure prospects for young scientists over this period.

We examined whether mandatory retirement explains this effect of team size and found that team size had a strong effect on career outcomes well before mandatory retirement. Mandatory retirement did increase the likelihood of exiting academia as well as reducing the likelihood of having a tenure-track job. However, mandatory retirement did not change the effect of team size on receiving federal funding, receiving tenure and leaving one's science field.

We also examined the association between the doubling of the NIH budget and team size. Indeed, in Fig. 2 we observed that the largest decline in academic positions occurred in life science fields. The NIH has raised concerns about the effect of funding being concentrated in the hands of few (likely older)

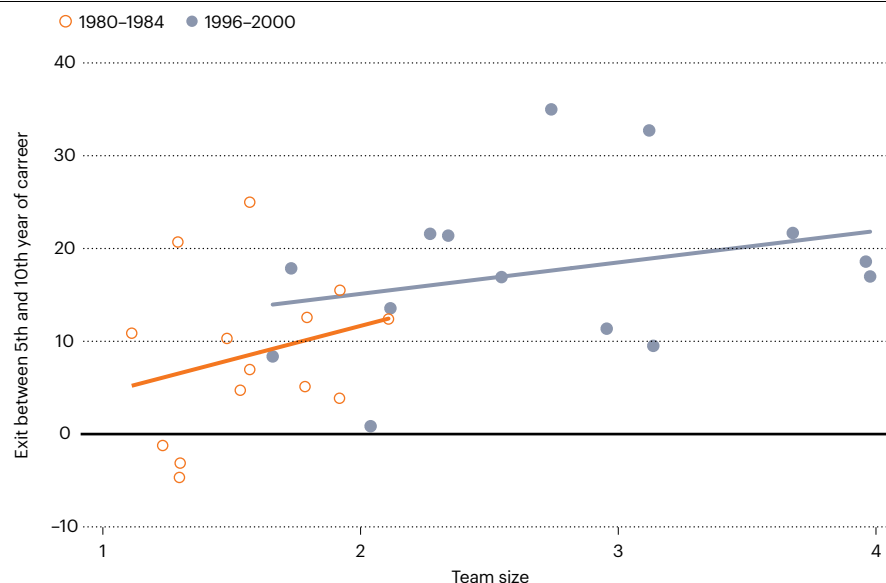


Fig. 4 | Exits from academia between 5th and 10th year of career by team size, before and after mandatory retirement. Sources: Survey of Doctorate Recipients and Web of Science.

scientists³¹. However, the focus was on the number of grants received by individual PIs and not on the size of labs.

There are theoretical explanations for why the careers of junior scientists might suffer when larger teams become the norm: primarily, it becomes more difficult for funding agencies and universities to be confident in the abilities of young scientists working in teams¹⁶. As a result, less funding goes to junior scientists and more funding and rewards end up being allocated to well-established scientists. The shift in funding away from junior scientists has led to concern that funding agencies may suffer from bias (such as the Matthew effect, in which young scientists' achievements are attributed to senior scientists on their team¹⁷), and we do find some evidence that female and foreign-born scientists have worse career outcomes. It has also led to policies that allocate some funding exclusively for young scientists. But it is possible that funding agencies are not highly biased and are mostly responding to the reduced information now available on younger scientists.

Some may argue that this is the result of an excess supply of graduate students relative to the demand for tenure-stream faculty. However, recent work demonstrates that 80% of faculty were trained at less than 28% of research universities³². Given this high concentration of hiring, if anything, the supply of doctorates from these elite institutions may outstrip the demand.

Future research should consider how NIH policies such as those designed to fund early-stage investigators or the K99/R00 early-career awards have impacted science careers and the total output of science. Although some prizes and awards exist to encourage junior researchers, these are often ex post facto and awarded only to the few, and schemes to identify and support junior researchers are unevenly spread across universities and research institutions. That said, understanding how team size affects information about individual research contributions is a topic for future research. Likewise, young scientists may want to diversify their research projects or work on smaller teams in order to highlight their research contributions. More generally, academic science has not adjusted its reward structure, which is largely individual, in response to a production technology that has become team based³³. Failing to address these concerns means a significant loss as junior scientists exit after a costly and specialized education in science, and/or fail to achieve their full potential because they face funding and promotion constraints.

Data availability

The Survey of Doctorate Recipients restricted use data are available by submitting an application to the National Center for Science and Engineering Statistics. Information on how to submit the application is available at <https://ncses.nsf.gov/about/licensing>.

Code availability

All code used to create the data and perform the analysis will be archived and made available to researchers who have obtained a license to the SDR data. Researchers interested in the code should contact D.K.G.

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Author contributions

All authors conceptualized the paper and participated in the writing of the original draft. M.A. and D.K.G. devised the methodology, investigated and visualized the data. C.d.F. and K.L. acquired the funding. C.d.F., K.L. and D.K.G. administrated the project. C.d.F. and D.K.G. supervised the project.

Competing interests

The authors declare no competing interests.

Additional information

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